

Teaching Accessibility: A Family Workshop for Middle School Students

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Abstract

Accessibility education is crucial for fostering inclusive technology design, yet it is often introduced at the higher education level, leaving younger students with limited exposure. This study explores how family-oriented learning can support accessibility awareness among middle school students and their families in the United States. We conducted an Accessibility Awareness Workshop within a U.S.-based STEM program, where participants engaged in educational games, videos, and discussions. Using pre- and post-surveys, observations, and discussions, we examined student learning, awareness shifts, and parental involvement. Findings suggest the workshop effectively fostered accessibility awareness and empathy, though engagement varied across activities. Challenges included attention variability, parent-child interdependence, and survey response fatigue. We propose design considerations for future accessibility education initiatives. This work provides insights into family-centered accessibility education, offering a foundation for more inclusive and engaging STEM learning experiences.

CCS Concepts

• **Human-centered computing** → **Accessibility**; • **Social and professional topics** → **K-12 education**; • **Applied computing** → **Interactive learning environments**.

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1 Introduction

Accessibility in computing education is essential for equipping the next generation of designers and developers with the knowledge needed to create inclusive digital environments. However, accessibility education is often introduced at the post-secondary level [13, 38], leaving younger students with little exposure to the concepts of inclusive design. Introducing accessibility early in education has the potential to normalize it as a core design consideration, rather than an afterthought. While prior research has explored accessibility education through classroom instruction and educational games [4, 29, 37], little work has examined how accessibility awareness can be fostered in family-oriented, non-classroom settings, particularly for underrepresented students in STEM.

To address this gap, we designed and conducted an Accessibility Awareness Workshop as part of a family-oriented U.S.-based STEM program for middle school students of color and their parents and school-aged siblings. The workshop introduced software accessibility concepts through a combination of interactive educational games, animated videos, and guided discussions. This setting allowed us to investigate how students engage with accessibility

concepts in a family learning environment, the impact of the workshop on their awareness of software accessibility, and the role of parental involvement in shaping their learning experience.

This exploratory study is guided by the following research questions:

- **RQ1:** *How do middle school students engage with and develop an understanding of accessibility concepts during a STEM workshop?*
- **RQ2:** *What is the role of parental involvement in shaping these learning experiences?*

2 Related Work

This section reviews literature on educational games for accessibility awareness and family co-learning models in STEM, particularly computing education, to contextualize our research.

2.1 Educational Games for Accessibility Awareness

Accessibility education is an integral part of computing education [36], ensuring that future software developers create inclusive software that meets the needs of users with disabilities. Accessibility topics are often covered in human-computer interaction (HCI) [25] and web design [37] courses, and even to non-computer science students [23].

Beyond traditional lectures and projects [4], educational games and gamification have emerged as engaging tools for fostering empathy and understanding of accessibility [29]. Wiesemüller et al. developed an immersive game-based learning experience for accessibility education in a virtual reality setting, leading to cognitive and affective learning outcomes [43]. Kletenik and Adler developed a set of accessibility simulation games to equip college students with awareness and knowledge of software accessibility [19–21]. Notably, Adler and Kletenik [1] suggested the suitability of these games for younger students, a notion supported by Zhou et al.’s successful implementation with grades 8–12 [45].

This trend aligns with broader research demonstrating the benefits of early exposure to computing concepts in K–12 [28]. Such exposure fosters foundational skills [44], enhances problem-solving abilities [28], and promotes long-term interest in computer science [3, 41], also encouraging participation from underrepresented groups [18, 35]. Therefore, extending accessibility education to younger students is a logical next step. Early learning can establish accessibility as a core design principle rather than an afterthought. Our work builds upon this idea, focusing specifically on middle school students from underrepresented backgrounds within a family-oriented learning environment.

2.2 Family Co-Learning Models in Education

Parental involvement has a well-established positive impact on children’s academic achievement [11], including in STEM fields [39]. This involvement has been linked to improvements in quantitative [14] and problem-solving skills [7], and increased student confidence [10].

The nature of parental involvement is influenced by socioeconomic status and cultural context. Higher socioeconomic status

often correlates with more direct parental support [24], while cultures like those of Chinese and Chinese Americans place strong emphasis on parental involvement in education, including decisions about school districts [8, 24].

Despite historical barriers to parental involvement, such as family structure, socioeconomic status, and parental education levels [42], research has shown diverse forms of engagement among Black families, including participation in parent groups, communication with school officials, school visits, and involvement in school decision-making [30]. Furthermore, parents and the extended community have been crucial in supporting Black students’ academic and social well-being [15].

2.2.1 Parental Involvement in Computing Education. Specifically within computing education, parental involvement in computational thinking has been extensively studied [6]. Homeschool parents play a key role in supporting their children’s understanding of computational thinking [12]. Studies using tools like ScratchJr have observed parents primarily taking on observer or coach roles, while children tend to be the planners [16]. Research also indicates that parental support and perceived usefulness enhance students’ motivation in visual programming [22], reinforcing earlier findings on parent-child collaboration in programming education [26].

Workshops have been effective in facilitating family co-learning in computing. Bresnihan et al. developed workshops to boost parental confidence and competence in engaging with computer science activities alongside their children [5]. Similarly, Cuellar’s robotics workshops demonstrated the positive impact of parental warmth and acceptance on children’s performance [9]. However, to our knowledge, there has been a lack of parent-child workshops focused on computer accessibility. Our study addresses this gap.

3 Methodology

3.1 Participants

This exploratory study was part of a U.S.-based family-oriented program that held family STEM nights engaging middle school students of color and their families in hands-on learning. The program introduced diverse engineering concepts while promoting family involvement and was conducted in collaboration with community organizations supporting underrepresented groups in STEM.

Our IRB-approved Accessibility Awareness Workshop was conducted during one of these evening sessions, leveraging the program’s established format to combine educational content with family engagement activities. Five families (14 participants) attended, with all student participants in grades 5–9. One parent chose not to complete surveys or activities but instead actively observed their child while still contributing to discussions. This setting allowed us to study how underrepresented middle school students learn about accessibility with their families.

3.2 Educational Materials

3.2.1 Games. The workshop utilized three educational games developed by Kletenik and Adler focused on building understanding of software accessibility through experiential learning [19–21]. The games employ a ball-popping mechanic while teaching concepts related to color blindness, deafness, and dyslexia. Engagement is

maintained through competitive gameplay against a computer opponent, creating natural opportunities for family discussion about accessibility between rounds.

The educational experience progresses through four distinct rounds. In *Round 1*, players begin with standard gameplay to establish familiarity. *Round 2* introduces simulated accessibility barriers—for instance, demonstrating how color vision differences impact gaming when color serves as the sole distinguishing feature between targets. *Round 3* and *Round 4* showcase the implementation and effectiveness of accessibility features, both in standard and simulated conditions.

Several enhancements were implemented to optimize the experience for middle school students. In a previous study, young participants became frustrated or confused after Round 2, as the increased difficulty led to game losses. The study also revealed that participants rarely had sufficient patience to read lengthy instruction text blocks [27].

To address the frustration and confusion that occurred after Round 2, we incorporated *supportive messages* explaining the corresponding accessibility barriers and reassuring students that it is okay to lose. These messages explicitly state that *Round 2* is intentionally designed to demonstrate the challenges faced by people with disabilities in inaccessible environments, helping students connect the experience to real-world accessibility issues in a constructive and encouraging way.

Moreover, the instructions were simplified with age-appropriate language and audio narration to support comprehension. By incorporating concise instructions and auditory guidance, we aimed to enhance engagement while ensuring students fully understood the learning objectives.

3.2.2 Videos. We developed animated videos featuring child characters in a classroom setting to introduce each accessibility topic. The decision to introduce animated videos explaining key terms and concepts was informed by prior research, which revealed that many students had limited knowledge of disabilities as well as key terminology such as “accessibility” and “dyslexia” [27] and did not often read text-based information. By presenting these topics through animations, we aimed to build foundational understanding about disabilities and accessibility before students played the games.

Each 2-minute video followed a similar narrative structure: introducing a student character with a disability, demonstrating a scenario they may encounter, and presenting inclusive strategies that enhance accessibility and support. For instance, the color blindness video featured a student who could not distinguish between red and green candies. The deafness video depicted a student who uses an ASL interpreter to communicate, while the dyslexia video showed a character struggling to understand the content written on the blackboard. We deliberately chose classroom settings and student-aged characters to create relatable scenarios for our middle school audience.

The educational narratives were crafted by researchers who translated accessibility concepts into age-appropriate scenarios and explanations for the middle school audience. The visual content was created using Animatron, a web-based animation platform, while ElevenLabs, a text-to-speech AI voice generator, was employed

solely for audio production to create more natural-sounding narration. These videos served to engage younger participants while providing context for the simulated disabilities in games and introducing corresponding assistive technologies (see Figure 1). We created three videos corresponding to the three games they played: color blindness, deafness, and dyslexia.



Figure 1: Video Screenshot - Color Blindness

3.3 Workshop Design

Our workshop was structured to teach basic software accessibility concepts to middle school students and their parents (see Figure 2).

The session began with a family dinner which served two purposes: allowing time for parental consent form completion and building a comfortable community atmosphere. During the dinner, we also introduced the project team, providing an opportunity for participants to become familiar with the researchers and the goals of the workshop. Following dinner, families were seated at tables equipped with laptops and headphones for the workshop.

Then, the educational activity was delivered. First, participants completed pre-surveys designed separately for students and parents to assess their initial understanding of accessibility. The workshop then opened with a slide presentation that connected STEM skills with accessible technology development. We established accessibility’s importance by highlighting both its broad impact, affecting people globally, and current implementation gaps in digital spaces. The presentation then connected these concepts with familiar examples of accessible design in everyday environments, such as building ramps and elevator braille.

The core workshop consisted of three sequential learning sessions, each following the same structure:

- (1) A short animated video providing child-friendly explanations of a disability and associated assistive tools.
- (2) Interactive gameplay designed to highlight accessible design principles for that disability.
- (3) A guided debrief about their experience.

After completing all game sessions, participants filled out post-surveys to measure changes in their understanding and gather feedback about the workshop experience. Finally, the workshop concluded with a comprehensive group discussion exploring overall impressions and suggestions for improvement.

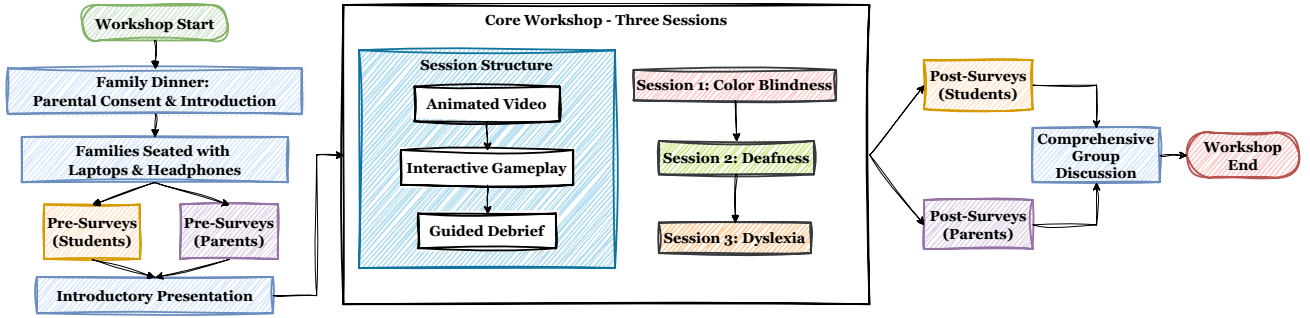


Figure 2: Workshop flow diagram showing the progression of activities.

3.4 Data Collection

We used open-ended and Likert-scale questions in pre- and post-surveys, designed separately for students and parents, to explore their understanding of accessibility and measure potential changes before and after our workshop. For a complete list of survey questions, please refer to the supplementary materials.

- **Student Surveys:** The pre-survey assessed prior knowledge of disabilities, accessibility and coding experience, while the post-survey measured learning gains and workshop feedback.
- **Parent Surveys:** The pre-survey evaluated familiarity with accessibility and confidence in discussing it with children, while the post-survey tracked changes in accessibility understanding and feedback.

Of the 14 participants, 8 were students and 6 were parents (see Table 1). All 8 students completed the pre-survey; however, one student’s responses appeared to be random and lacking seriousness, so their data was excluded from the analysis, leaving a total of 7 valid student responses. Only 4 students completed the post-survey.

Among the 6 parents, 5 completed the pre-survey (two from the same family and three from different families). One parent from the fifth family opted not to complete either survey. Additionally, one parent who completed the pre-survey did not complete the post-survey, since their family arrived late to the workshop.

Participant Group	Pre-Survey Completed	Post-Survey Completed
Students (Total: 8)	8 (1 excluded, valid: 7)	4
Parents (Total: 6)	5	4

Table 1: Survey completion rates for students and parents.

In addition to survey data, two note-takers observed the entire session, including the survey process, slide presentation, gameplay interactions, debrief discussions, and final group discussions. These observation and discussion notes provided qualitative insights into participant engagement, learning behaviors, and parent-child interactions throughout the workshop.

3.5 Data Analysis

To analyze qualitative data collected from participant responses, we conducted a thematic analysis following the step-by-step guide outlined by Maguire and Delahunt [32]. The themes from the thematic analysis were organized into two groups, each group corresponding to one of the research questions guiding the study. Three themes, “Active Engagement in Interactive Learning,” “Changes in Accessibility Awareness,” and “Challenges in Maintaining Attention,” were categorized under the first research question (RQ1). Three themes, “Collaborative Learning,” “Engagement Interdependence,” and “Parent Awareness,” related to the second research question (RQ2). We supplement these themes with pre- and post-survey results, using quantitative responses to support observed trends.

3.6 Ethical Considerations

This study was approved by the Institutional Review Board (IRB), ensuring ethical compliance. Parental consent was obtained at the entrance, where parents signed written consent forms upon arrival. For student participants, the pre-survey included a detailed project description, and the first question explicitly asked, “Are you willing to participate?” to confirm their assent before proceeding. Participants were informed about the study’s purpose, procedures, and their right to withdraw at any time without consequence.

4 Preliminary Findings

4.1 RQ1: Student Engagement and Understanding of Accessibility

4.1.1 Active Engagement in Interactive Learning. The interactive games elicited high engagement, with students demonstrating active participation and understanding of the educational purpose. For example, one student explained to their parent that the second round of the game was intentionally more difficult to simulate real-world accessibility barriers and the lack of assistive solutions. This kind of reflection indicates that students were not only experiencing the challenges but also internalizing accessibility principles.

The animated videos also proved effective in supporting student understanding. Observers noted active listening behaviors such as nodding, particularly when videos introduced specific disabilities and assistive technologies. For instance, in the deafness session, the video introduced how Deaf and hard-of-hearing individuals communicate using American Sign Language (ASL). A student

with prior ASL knowledge pointed out that sign language varies internationally, rather than being universal.

4.1.2 Changes in Accessibility Awareness. Pre-survey responses indicated a limited understanding of accessibility among students at the start of the workshop. Despite many students (4/7) having heard of accessibility in school, only one could accurately explain what it means for a building and website to be accessible when asked about both contexts in the pre-survey. Additionally, only two students were able to name any assistive technologies designed to help individuals with disabilities. Similarly, when prompted with the question, “*Imagine someone is designing a game and wants to make sure everyone can play it. Who should they ask to test it?*,” only two students included people with disabilities in their answer.

Through hands-on experiences, students began developing a deeper understanding. During the color blindness game debrief, one student proposed potential accessibility solutions such as audio cues and colorblind glasses, demonstrating an increased awareness of accessibility challenges and a desire to find solutions. This suggests that students were not only learning about accessibility but also beginning to think critically about inclusive design. Additionally, in the final discussion session, one student reflected on the importance of experiencing accessibility barriers firsthand, stating: “*putting yourself in the shoes [of others] creates more empathy, so it means a bit more to us.*” This response suggests that first-hand experience played a key role in fostering empathy and deeper understanding of accessibility issues.

Post-survey responses, though fewer in number, showed notable improvements. One student who had initially showed uncertainty about the meaning of “accessible websites” later wrote: “*I believe that website accessibility is to make it work for more types of people with disabilities.*”

Also, one student, when asked in the pre-survey about who they should ask to test a game, had initially mentioned only “coding experts.” In the post-survey, their response changed to “People who have disabilities,” demonstrating a clear shift toward recognizing the importance of inclusive design.

4.1.3 Challenges in Maintaining Attention. The non-classroom environment introduced unique challenges. Despite audio support options, some students skimmed through instructions. Moreover, some students rushed into gameplay without reading the guidelines, temporarily left the activity area, or arrived late, which sometimes disrupted the intended learning sequence. Such behaviors demonstrate the challenges of maintaining structured learning activities in informal, family-based environments.

Additionally, some students exhibited low engagement during the survey process. Several provided brief, non-informative responses, and fewer students completed post-surveys suggesting possible survey fatigue. Two key factors likely contributed to this:

- (1) The overall length of the surveys may have been overwhelming for students, particularly after engaging in multiple game sessions.
- (2) The open-ended questions required students to provide detailed responses, which many seemed unwilling to do or caused survey fatigue.

These observations underscore the need for more streamlined transitions between activities and a more age-appropriate survey design.

4.2 RQ2: Role of Parental Involvement

4.2.1 Collaborative Learning. Parental involvement played a significant role in shaping the workshop experience. Many parents actively participated in discussions and engaged with their children during activities. Some families naturally transformed the games into friendly competitions, creating an atmosphere of enjoyable learning marked by frequent laughter and visible enjoyment during sessions. These moments highlight the potential of family-oriented workshops to create enjoyable and interactive learning environments.

Parents also made meaningful contributions to discussions. Several actively shared personal experiences, such as a parent with dyslexia who contextualized accessibility challenges by explaining how they navigated similar barriers in daily life. This kind of personal connection not only enriched the learning experience for students but also underscored the real-world impact of accessibility issues. Some parents frequently asked questions about assistive technologies and accessibility tools, indicating a high level of engagement with the workshop content. For example, after watching the video on dyslexia, one parent asked about a tool mentioned for assisting with reading webpages and expressed interest in trying it.

4.2.2 Engagement Interdependence. While some parents were highly engaged, others took a more passive role, affecting the overall learning experience. For example, some participants lost focus during slides. This variability indicates a need for more interactive and discussion-based approaches that encourage continuous participation from both parents and students.

Observations revealed a strong connection between parental engagement and student participation. When parents were actively engaged, their children exhibited higher levels of participation and curiosity. Conversely, when parents became distracted—often checking their phones or disengaging during presentations—students’ engagement also declined. These findings highlight the importance of structured parent-child learning interactions in family-oriented workshops.

4.2.3 Parent Awareness. Although our research centered on middle school students, the workshop design revealed an unexpected outcome in parents’ evolving awareness.

During the final group discussion, parents shared reflections that revealed transformations in their perspectives through the workshop experience. Several parents expressed that their views had changed, moving from seeing accessibility as an abstract concern to understanding its practical implications. One parent reflected that they had previously taken the basic abilities demonstrated in the accessibility games for granted. However, after participating in the game sessions, they recognized the importance of accessible design in addressing barriers.

The shift in perception extended beyond the workshop setting into workplace contexts. This transformation was exemplified by one parent who worked in higher education, who reflected on Title II of the Americans with Disabilities Act. Their understanding

evolved from seeing it as a workplace requirement to appreciating its crucial role in ensuring digital accessibility, particularly relevant given the Department of Justice’s April 2024 final rule requiring accessible websites and mobile applications in state and local government institutions.

In addition, post-survey responses indicate that parents felt more confident discussing accessibility with their children after the workshop. In response to the question “*How confident do you feel explaining the concept of accessibility to your middle school child?*”, the average confidence rating increased from 3.4 (pre-survey) to 4.5 (post-survey).

These findings suggest that the workshop was successful in fostering a broader understanding of accessibility, not only among students but also among parents who can further influence their children’s learning and advocacy for inclusive practices.

5 Discussion

Our workshop extends the growing literature on family-oriented approaches to STEM education [34, 39, 40], which includes strategies for broadening participation among diverse populations, with students of color being one important focus area. The findings from this exploratory study provide key insights into how accessibility concepts can be effectively introduced to middle school students in a family-oriented, non-classroom setting. The results demonstrate both the potential and challenges of engaging families in accessibility education through interactive approaches.

The combination of accessibility games and animated videos proved effective in fostering understanding of accessibility concepts. This aligns with prior research showing that game-based learning methods and animated visual content enhance student engagement and conceptual understanding [2, 17, 31]. Students demonstrated increased empathy and awareness through hands-on experiences, particularly evident in their ability to propose accessibility solutions after experiencing simulated barriers. The workshop structure effectively facilitated knowledge transfer, as evidenced by the pre- and post-survey comparisons that demonstrate shifts in student perspectives on accessibility.

Parent involvement emerged as a critical factor in the learning process. Prior studies have shown that parental-engaged learning environments contribute to increased younger students’ learning experience, shaping student engagement and fostering collaboration and interaction in STEM education [6, 22, 26, 30, 33, 39]. When actively engaged, parents enriched discussions with personal experiences and workplace connections, helping bridge theoretical concepts with real-world practices. Post-survey results indicate a notable increase in parental confidence. This suggests that participating in discussions and experiencing accessibility-related challenges firsthand helped parents feel better equipped to engage in conversations about accessibility with their child.

Despite these successes, the non-classroom environment presented unique engagement challenges. Unlike traditional classroom settings where structured routines and teacher-led facilitation guide attention, the family-oriented workshop allowed for more fluid participation, leading to variability in participant focus. The introductory presentation failed to capture consistent attention. Some students skimmed the slides and immediately started playing the

games, bypassing the presentation, videos, and debrief discussions. This suggests that with a family-oriented format, elements typically associated with traditional classroom delivery, such as lecture-style presentations and text-heavy instructions, need to be redesigned into more interactive formats.

5.1 Implications for Teaching Accessibility and Designing Family-Oriented Workshops

First, from the survey responses, we noticed a disconnect between parent and student perceptions of accessibility discussions at home. While 4 out of 5 parents reported having talked about “disabilities” or “accessibility” with their children, only 1 student recalled hearing these topics discussed at home. This suggests potential gaps in effective communication about accessibility between parents and children, or differences in how parents and children perceive and remember these discussions. Although we do not have direct mappings between specific parents and students, these findings indicate that students may not always recognize the discussions about disabilities or accessibility. This is similar to Zhou et al. who found that while some teachers reported teaching accessibility, most students reported not having learned accessibility in their courses [45]. It is possible that while parents believe they have introduced these topics, their child may not fully recognize or remember these discussions as explicitly addressing “accessibility” or “disabilities.” More work needs to explicitly consider the topic of accessibility.

Next, our experience suggests several key considerations for designing family-oriented computing workshops, elaborated below.

- (1) **Content Structure and Delivery:** To improve engagement, future workshops should incorporate clearer transitions between activities and reduce reliance on text-heavy content. Using more visual elements and shorter, focused instructional segments can help sustain attention, especially for younger students. Embedding instructional videos directly within the game platform could also allow for more flexible, self-paced learning.
- (2) **Parent-Child Interaction:** Given the strong influence of parental engagement, future workshops should actively encourage parent-child collaboration through structured joint activities. To foster student independence, parallel but complementary tasks for parents and children could allow both to engage meaningfully while ensuring students develop autonomy in their learning process.
- (3) **Environment and Facilitation:** While the informal setting supported organic learning, it also led to engagement variability. A clearer agenda and structured facilitation could help maintain focus while still preserving family-oriented flexibility.
- (4) **Survey Structure:** Survey fatigue led to incomplete responses, particularly for open-ended questions. Future studies should consider administering only a post-survey for younger students, reducing the number of questions, and using multiple-choice or scaled formats to improve response quality while minimizing cognitive load.

5.2 Limitation

While this study provides valuable insights into family-oriented accessibility education, several limitations should be considered when interpreting the findings. First, the sample size was relatively small, consisting of only five families with 14 participants. Further, a longitudinal study with multiple workshop sessions or follow-up activities could provide better insights.

6 Conclusion

Our study demonstrates both the potential and challenges of teaching accessibility concepts through family-oriented workshops. By integrating educational games, animated videos, and structured discussions, the workshop provided opportunities for students to engage with accessibility concepts, reflect on real-world challenges, and explore potential accessible solutions. Parental involvement played a critical role, with active participation enhancing student learning.

While students showed increased awareness and empathy after engaging the workshop, engagement fluctuated. Though limited in scope, this study provides qualitative insights into family-centered accessibility education, laying a foundation for future work to explore more interactive presentation formats, structured parent-child activities, and implementations in other STEM education programs.

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